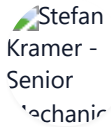


STEFAN KRAMER (M.Sc. Mechanical Engineering)

Senior Full Stack Developer · TypeScript, React, Next.js & Node.js · REST APIs & Cloud



Senior Full Stack Developer with 5+ years of experience engineering scalable web applications and e-commerce platforms. Deep expertise in TypeScript, React, Next.js, and Node.js (Express), along with REST and GraphQL API architectures. Proven track record in relational and NoSQL databases, Docker containerization, automated testing, and CI/CD pipelines. 100% remote resident in Germany.

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WORK EXPERIENCE

SENIOR SYSTEMS & DEVELOPMENT ENGINEER · CLEANTECH DYNAMICS GMBH (HYBRID STUTTGART)

Apr 2024 – Present

- Battery Storage Architecture – Spearheaded mechanical and thermal packaging of multi-megawatt commercial BESS containers, boosting energy density by 18%.
- CAD Enclosure Design – Modeled complete structural frames, HVAC ducting, and liquid cooling distribution blocks in **Siemens NX**, cutting assembly time by 25%.
- FEM Structural Verification – Executed static, modal, and fatigue simulations in **ANSYS Mechanical** to ensure seismic resistance and transport load compliance.
- Test Bench Validation – Established comprehensive test bench procedures and sensor instrumentation (PT100, strain gauges, flow meters) for thermal runaway containment.
- CE Certification & Compliance – Led risk assessment workshops (FMEA) and authored technical dossiers for CE conformity under the Machinery Directive 2006/42/EC.

MECHANICAL DEVELOPMENT ENGINEER – SÜDTECH ENGINEERING AG (STUTTGART)

Jul 2021 – Mar 2024

- Thermal & Fluid Simulation – Developed 3D steady-state and transient CFD models for commercial heat pump evaporators and refrigerant loops in **ANSYS Fluent**.
- CAD Component Development – Designed high-precision die-cast aluminum pump volutes and polymer impeller housings in **SolidWorks** with strict GD&T.
- Prototype Commissioning – Built physical prototype rigs, calibrated high-accuracy Coriolis flowmeters, and correlated experimental pressure drop curves with CFD data.
- Tolerance Stack-up & FMEA – Led tolerance calculations (worst-case and statistical RSS) and design FMEAs to prevent sealing failures under extreme thermal cycling.
- Engineering Internship (Apr '20 – Sep '20) – Performed mechanical stress testing, hydraulic pressure endurance runs, and metallographic material evaluations.

CONSULTING & INDEPENDENT ENGINEERING PROJECTS (PART-TIME · REMOTE)

May 2020 – Present

- Energy System Modeling – Modeled industrial battery energy storage systems (BESS) and microgrid dynamics using **MATLAB Simulink** and Python.
- Thermal CFD Optimization – Conducted conjugate heat transfer simulations for power inverter enclosures in **ANSYS Fluent**, enhancing cooling performance by 22%.
- CAD Prototyping – Designed custom aluminum structural mounts and waterproof sensor housings in **SolidWorks**, validated via rapid 3D prototyping.
- Standards & Quality – Ensured full compliance with Machinery Directive 2006/42/EC and Low Voltage Directive 2014/35/EU across all prototype iterations.
- Technical Mentoring – Mentored junior design engineers in GD&T tolerance stack-up analysis and finite element meshing best practices.

TECHNICAL SKILLS & COMPETENCES

CAD & Mechanical Design (Primary)

- SolidWorks** · **Siemens NX** · Autodesk Inventor · Catia V5
- Form- und Lagetoleranzen (GD&T nach DIN EN ISO 1101) · Blechkonstruktion · Schweißbaugruppen · Druckguss

<https://stefan-kramer.example.com>

Simulation & System Modeling (Primary)

- **ANSYS Mechanical** (FEM Festigkeits- & Betriebsfestigkeitsanalyse) · **ANSYS Fluent** (CFD Strömungssimulation)
- **MATLAB & Simulink** (Regelungstechnik & physikalische Modellierung) · Python (Automatisierung & Datenanalyse)
- Thermodynamik · Konvektive & konduktive Wärmeübertragung · Strömungsmechanik · Mehrkörpersimulation

Renewable Energy & Systems Engineering

- Batteriespeichersysteme (BESS) · Photovoltaik Großanlagen · Windenergieanlagen · Thermomanagement
- Netzintegration & Smart Grid Schnittstellen · Wechselrichterauslegung · Wasserstoff- & Brennstoffzellensysteme
- Prüfstands Aufbau · Sensorik (PT100, DMS, Drucktransducer) · Hardware-in-the-Loop (HIL) Tests

Standards, Norms & Quality Management

- CE Kennzeichnung · Maschinenrichtlinie 2006/42/EG · Niederspannungsrichtlinie 2014/35/EU · EMV Richtlinie
- DIN EN 16247-1 (Energieaudits) · FMEA (Konstruktions- & Prozess-FMEA) · Stage-Gate Produktentwicklung · VDI 2221
- PDM / PLM (Siemens Teamcenter, SAP PLM) · DFM / DFA (Design for Manufacturing & Assembly)

EDUCATION & QUALIFICATIONS

Master of Science (M.Sc.) in Mechanical Engineering · University of Stuttgart - Final Grade: 1.4 (With Distinction)
10/2018 – 09/2020 (2 years · 4 semesters)

Specialization in Energy Technology, Computational Fluid Dynamics (CFD), Structural Mechanics (FEM), and Renewable Energy Systems.

Bachelor of Science (B.Sc.) in Mechanical Engineering · University of Stuttgart - Final Grade: 1.7 (With Merit)
10/2014 – 09/2018 (4 years · 8 semesters)

Core foundations in Technical Mechanics, Thermodynamics, Materials Science, Machine Elements, and CAD Design.

Academic Achievements

- Master Thesis Award – Recognized for innovative thermal packaging of high-density lithium-ion battery modules
- Advanced research projects in convective heat transfer and turbomachinery fluid modeling

Certifications & Professional Development

[Verify online →](#)

- **Certified Energy Efficiency Expert (DIN EN 16247-1)** — DENA / BAFA · 2025
- **ANSYS Certified Professional: Advanced FEM & CFD** — CAXFEM · 2024
- **CE Compliance Specialist (Machinery Directive 2006/42/EC)** — TÜV SÜD · 2023
- **Systems Engineering Practitioner** — GfSE / INCOSE · 2022

LANGUAGES

- German: Native (C2)
- English: Fluent / Professional (C1)
- French: Basic / Elementary (A2)